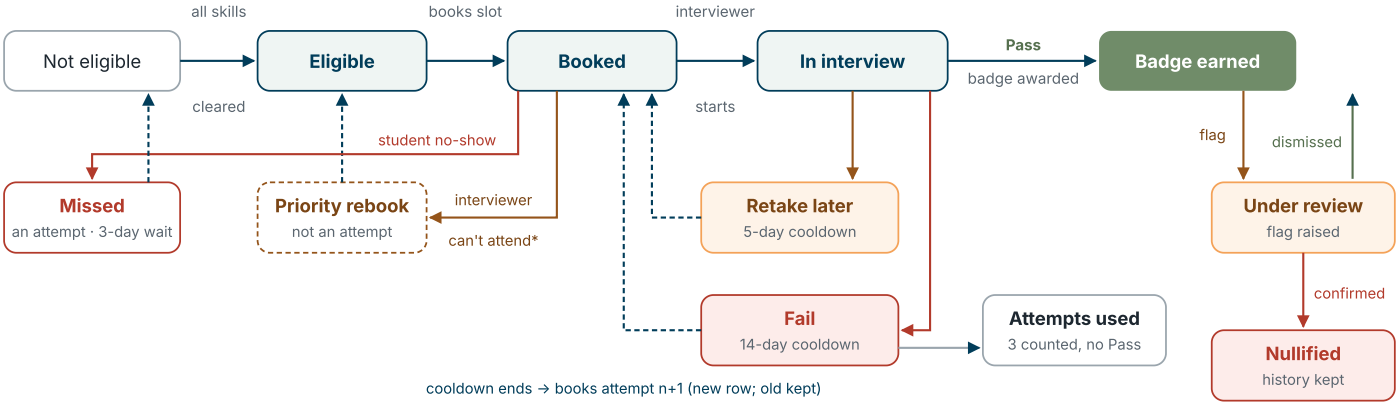
GRIT Interview Hub — Interview Scheduling & Badge Lifecycle

AI Product Engineer · Round 1 · Abhinav Chaudhary
November 2026 · prototype + Supabase/PostgreSQL backend

One journey per student × badge level (Novice, then Master), because each level has its own interview. Each dimension is stored separately: eligibility (computed from skills, never a flag), booking, attendance, interview, outcome, badge and malpractice case. The state the student sees is **derived** from those, so it can never contradict them.

1 Student interview lifecycle — state transitions



* first offered to a colleague free at the same time; only if none is free does it become a priority rebook. Dashed = back into the booking loop. If the slot ends with no outcome the student sees "Awaiting result". Master unlocks only while Novice is held.

2 Key transition triggers

Transition	Trigger (who)	Guard enforced by the database
Not eligible → Eligible	last required skill cleared (contest system)	`is_eligible()` over skill rows + previous level held; students cannot clear their own skills
Eligible → Booked	student picks a slot	owner check, cooldown, attempt limit, ≥ 2 h notice, ≤ 21 days ahead, interviewer active, no time clash
Booked → Booked / Eligible	reschedule (max 2) / cancel (student or ops)	students stop 12 h before; ops override is audited; reschedule keeps the same attempt
Booked → In interview	assigned interviewer starts it	only that interviewer; from 60 min before start; attendance = present
In interview → Pass / Retake / Fail	assigned interviewer submits	once only; scores 1–5; ≥ 20-char feedback for non-Pass; Pass awards the badge in the same transaction
Result corrected	interviewer ≤ 24 h, or ops	reason required, revision row kept; a Pass is never reversed here — only through review
Earned → Under review → Nullified / Earned	flag (ops, or the student's interviewer) → reviewer decides	reviewer permission; confirm only after an investigation; written reason; back to Earned only when no other case is open

3 Important edge cases

Case	How it is handled
Two students, same slot, same moment	Row lock on the slot + unique index (1 active booking per slot). Loser: "someone just booked that slot". Proven with two live DB sessions.
Double-click / network retry	Idempotency key returns the original booking; advisory lock blocks two different requests for the same student × badge.
Interviewer removes availability or can't attend	Removing a booked window is refused. "I can't make it" reassigns to a free colleague, else priority rebook (never the student's attempt).
Interviewer forgets the outcome	Student sees "Awaiting result"; ops get a pending list + reminder; outcome can still be submitted.
Skill requirements change after booking	Booked students keep the slot (grandfathered, snapshot timestamp); unbooked students are re-evaluated instantly.
Malpractice after a Pass · two flags on one badge	A flag only moves the badge to review. Dismissing one case keeps it under review while another is open. Students never see evidence.
Notification delivery fails	Notification is queued/marked failed; the booking or result still stands (delivery is decoupled).

Assumptions, data model & governance

4 Product assumptions

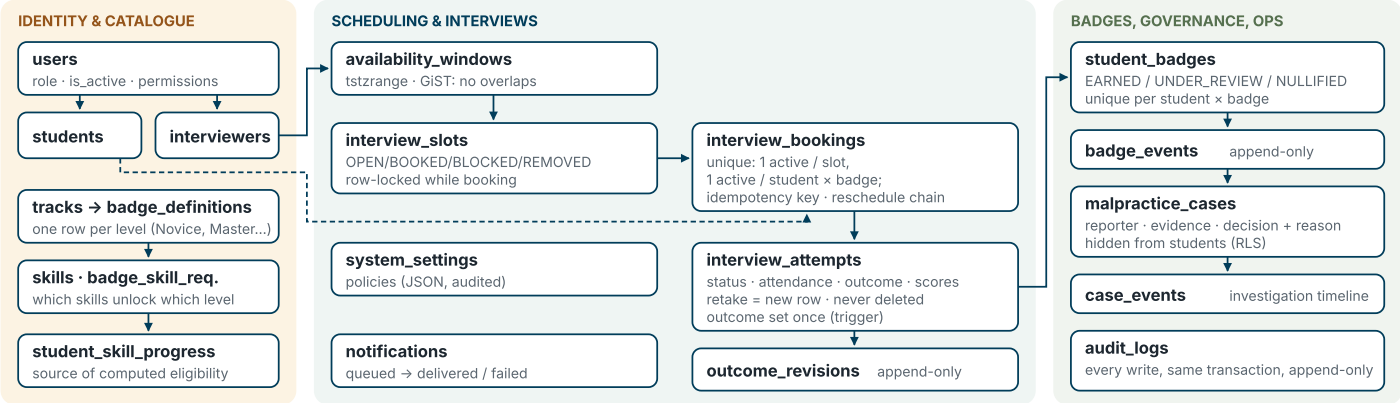
GIVEN BY THE BRIEF

- Interview unlocks when every required skill for the Track Badge is cleared.
- Small interviewer team, each with own availability; students book from their dashboard.
- Outcomes: Pass, Fail, Retake later. Pass earns the badge.
- Ops see which eligible students have booked and which have not.
- A badge can be nullified if malpractice is flagged later.

MY ASSUMPTIONS — all editable by ops, all audited

- 30-min interviews, 10-min buffer; book 2 h–21 days ahead; change until 12 h before (max 2 reschedules).
- Retake later = 5-day cooldown (mirrors GRIT's 5-day re-attempt gap); Fail = 14 days; 3 counted attempts per badge.
- Student no-show counts as an attempt (3-day cooldown); interviewer no-show or cancellation never does.
- Interviewers correct non-Pass results within 24 h; ops after. Only reviewers decide malpractice.
- **Open questions:** Fail vs Retake semantics · attempt limit · appeal window · does a nullified Novice void Master?

5 Database schema & entity relationships — PostgreSQL / Supabase, 3 migrations, 20 tables



6 Booking concurrency protection

- **Serialise:** `book_slot()` takes an advisory lock per student × badge, then `SELECT … FOR UPDATE` on the slot — concurrent requests queue, then re-check status.
- **Re-validate inside the lock:** eligibility, cooldown, attempts, notice, horizon, interviewer active, the student's other bookings.
- **Backstop:** partial unique indexes (one active booking per slot; per student × badge) and GiST exclusion (no overlapping slots per interviewer).
- **Idempotent:** a retried request with the same key returns the original booking. Reschedules lock both slots in id order — no deadlocks.
- **Atomic:** slot, booking, attempt, audit and notification commit together or not at all.

7 Malpractice & badge governance

- **Flag ≠ decision.** Ops, or an interviewer who interviewed that student, may flag. The badge moves to *Under review* and stays visible to the student, with neutral wording.
- **Decide:** only reviewers (`can_review_malpractice`). Confirm requires an open investigation and a ≥ 15-char reason; the student is told the outcome, never the evidence.
- **Never delete:** nullification is a status change; `earned_at`, badge events, case timeline and audit rows are append-only (triggers).
- **Defence in depth:** anon has no access; signed-in users only SELECT through RLS; every write is a `SECURITY DEFINER` RPC that checks the caller from `auth.uid()`, with pinned `search_path`.

Verified: 41 domain tests · 94 database checks incl. RLS and privilege sweeps · 2 live two-session booking races · 42 browser checks across student, interviewer, ops, malpractice and mobile.